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Introduction to use TASK

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1. Basic operation
2. How to setting parameters
3. Operation of equilibrium module: TASK/EQ
4. Operation of transport module: TASK/TR
5. Coupled operation of TASK/EQ and TASK/TR

Basic operation

- **Setup of graphic library GSAF**
 - **At the beginning of TASK codes, setup of GSAF is required.**
 - **Select graphic resolution** (0: metafile only, no graphics, n:)
 - **Input one character command**
 - c: continue
 - f: set metafile name (e.g. xxx.gs) and start saving
- **Choice of graphic operation**
 - **At the end of one page drawing, choose one char. command**
 - **c** or **CR**: change focus to original window and continue
 - **f**: set metafile name and start saving
 - **s**: start saving and save this page
 - **y**: save this page and continue
 - **n**: continue without saving
 - **d**: dump this page as a bitmap file “gsdumpn”
 - **q**: quit program after confirmation

Graphic Utilities

- **Utility program**

- **gsview**: View metafile
- **gsprint**: Print metafile on a postscript printer
- **gstoeps**: Convert metafile to eps files of each page
- **gstops**: Convert metafile to a postscript file of all pages

- **Options**

- **-a**: output all pages, otherwise interactive mode
- **-s ps**: output from page ps
- **-e pe**: output until page pe
- **-p np**: output contiguous *np* pages on a sheet
- **-b**: output without title

- **Example**

- **gstops -ab xxx.gs**: convert all figures to one postscript file
- **gstoeps -ab xxx.gs**: convert each figure to a eps file

Typical File Name of TASK

- **XXcomm.f90**: Definition of global variables, allocation of arrays
- **XXmain.f90**: Main program for standalone use, read XXparm file
- **XXinit.f90**: Initialize input parameters
- **XXmenu.f90**: Command input
 - **XXparm.f90**: Read input parameters
 - **XXview.f90**: Show input parameters
 - **XXprep.f90**: Initialization of run, initial profile
 - **XXexec.f90**: Execution of run
 - **XXgout.f90**: Graphic output
 - **XXfout.f90**: Text file output
 - **XXsave.f90**: Binary file output
 - **XXload.f90**: Binary file input

Typical input command

- When input line includes **=**, interpreted as a namelist input (e.g., **rr=6.5**)
- When the first character is not an alphabet, interpreted as a one-character command
 - **r**: Initialize profiles and execute
 - **c**: Continue run
 - **p**: Namelist input of input parameters
 - **v**: Display of input parameters
 - **g**: Graphic output
 - **w**: print output
 - **s**: Save results into a file
 - **l**: Load results from a file
 - **q**: End of the program
- When the first character is '!', the line is neglected

How to setting input parameters

- **Default setting of the module**

- Default parameters are set at the subroutine `XX_init` in `XXinit.f90`

- **Preset parameter file**

- If there is a namelist file `XXparm` in the executing directory, the module reads the file after the default parameter setting in `XX_init`.
- Be careful not to leave an unnecessary file `XXparm` for avoiding unintentional set of parameters.

- **Setup of parameters by input lines**

- You can set parameters by an input line of the namelist form
`name1=value1, name2=value2, name3=value3`
- You can set a list of parameters by the input lines after the one char. command “p” in the form of namelist file

`&XX`

`name1=value1, name2=value2, name3=value3` (more lines)

`/`

How to run TASK/EQ (1)

- **Interactive operation with default parameters**
 - `cd task/eq`
 - If there is a file named `eqparm` in this directory, remove or rename it
 - On macOS, start the module from a `xterm` window, not from a terminal window.
 - Key input sequence
 - `./eq` (start eq module)
 - `5` (window size 1024x760)
 - `c` (continue operation)
 - `r` (run the module with default setting)
 - `g` (start eq graphic interface)
 - `s2` (2D standard plot)
 - `CR,CR,CR,CR` (repeat Carriage Return 4 times)
 - `x` (exit eq graphic interface)
 - `q` (quit eq module)

How to run TASK/EQ (2)

- **Interactive operation with a preset file**
 - `cd task/eq`
 - `cp parm/eqparm.ITER eqparm`
 - On macOS, start the module from a **xterm** window, not from a terminal window.
 - Key input sequence
 - `./eq` (start eq module)
 - `5` (window size 1024x760)
 - `c` (continue operation)
 - `r` (run the module with default setting)
 - `g` (start eq graphic interface)
 - `s` (1d and 2D standard plot)
 - `CR×10` (repeat Carriage Return 10 times)
 - `x` (exit eq graphic interface)
 - `q` (quit eq module)
- `rm eqparm` (remove eqparm file)

How to run TASK/EQ (3)

- **Batch operation with an input file**

- `cd task/eq`
- Start the module which reads an existing input file (`in/eq.ITER01.in`), writes an output file (`eq.ITER01.out`), and generates a graphic metafile (`eq.ITER01.gs`)
`./eq <in/eq.ITER01.in >eq.ITER01.out`
- View the graphic output
On macOS, start from a `xterm` window, not from a terminal window.
`gsview eq.ITER01.gs`
Enter figure page number or 0 for all
- Convert the graphic metafile to a postscript file
`gstops -ab eq.ITER01.gs >eq.ITER01.ps`
- Convert all the figures in the graphic metafile to EPS files
`gstoeeps -ab eq.ITER01.gs`

How to run TASK/TR (1)

- **Interactive operation with default parameters**

- Example of key input sequence

cd task/tr

./tr2 (start tr module: tr may conflict with a unix command)

5 (window size 1024x760)

c (continue operation)

r (start a run with default setting)

g (start tr graphic interface)

t6 (time evolution) CR for next graphic input

r1 (radial profile)

x (exit tr graphic interface)

c (continue the run)

g (start tr graphic interface)

t6 (time evolution)

r1 (radial profile)

x (exit tr graphic interface)

q (quit eq module)

How to run TASK/TR (2)

- **Interactive operation with a preset file**
 - `cd task/eq`
 - `cp parm/trparm.ITER trparm`
 - `./tr2` (start tr module)
 - 5 (window size 1024x760)
 - c (continue operation)
 - r (run the module with default setting)
 - `ntmax=1000 nbtot=25` (change parameters)
 - c (continue run)
 - g (start eq graphic interface at 6 s)
 - t6 (time evolution, CR for next input)
 - r1 (radial profile)
 - x (exit eq graphic interface)
 - q (quit eq module)
 - `rm trparm` (remove trparm file)

How to run TASK/TR (3)

- **Batch operation with an input file**

- Input file: `in/tr.ITER01.in`
- This example uses a fixed-boundary equilibrium file `../eq/eqdata.ITER01` in a previous eq run
- `cd task/tr`
- `./tr2 <in/tr.ITER01.in | tee tr.ITER01.out`
- `gsview tr.ITER01.gs`